

CASE STUDY: China-Linked SIM-Based DDoS Attack Targeting New York City Telecommunications

1. Introduction

Distributed Denial of Service (DDoS) attacks have traditionally relied on botnets of infected computers. However, attackers are developing new and more powerful methods that use unconventional devices such as **IoT systems**, compromised mobile phones, and SIM card farms.

This case study examines an incident described by a cybersecurity expert who analysed a China linked cyber tool designed to flood a major U.S. telecom station with massive traffic generated through SIM cards. If the attack had succeeded, it could have caused telecommunications outages across parts of New York City, disrupting emergency services, financial institutions, and critical infrastructure.

2. Background

Telecommunications networks process enormous volumes of traffic every day, and they normally use redundancy and filtering to manage the load.

The attack method in this case bypassed traditional DDoS protections by using:

- Large numbers of SIM based authentication requests
- Traffic that appeared to come from legitimate cellular networks
- Possibly compromised base stations, SIM farms, or cloned SIM cards
- Spoofed IMEI and IMSI identifiers
- Distribution across thousands of mobile devices

Traditional DDoS systems detect suspicious devices or IP networks. In this situation, the attack traffic looked like normal subscriber traffic, so detection was extremely difficult.

3. Attack Overview

The attackers coordinated thousands or possibly millions of SIM cards or devices pretending to be SIM cards. These devices generated a flood of:

- Signaling requests
- Authentication messages
- Registration attempts
- Location update requests

All of this traffic targeted a telecom switching station that serves the New York City region.

When mobile networks receive these requests, they must process them before a device can connect. Overloading this system can result in:

- Network registration failure

- SMS and call disruptions
- Mobile data outages
- Inability to reach 911

This category of attack is known as:

- SS7 and Diameter Signaling DDoS
- SIM Based DDoS Attack

These attacks target the control plane of the network. The control plane handles authentication and routing. It has limited capacity compared to the data plane and is easier to overload.

4. Attack Chain Breakdown

4.1 Stage 1: Reconnaissance

Attackers likely researched:

- Which telecom nodes serve New York City
- Which authentication servers provide the most value
- Which signaling systems have weak rate limits
- How to simulate subscriber identity without triggering alerts

China linked groups often perform long term reconnaissance before launching operations.

4.2 Stage 2: SIM Exploitation

The attackers used one or more of the following methods:

- SIM farms containing thousands of SIMs in modem banks
- SIM cloning
- Virtual SIMs generated by custom hardware
- Compromised IoT devices with embedded SIMs

These devices repeatedly sent:

- Attach requests
- Authentication packets
- Tower handshake messages
- Location update messages

4.3 Stage 3: Signaling Flood

The devices then sent a massive wave of requests toward a specific telecom switching centre.

This overwhelmed the Mobile Switching Center (MSC) and Home Location Register (HLR), which could not handle the volume.

The traffic:

- Appeared legitimate
- Came from global mobile carriers
- Used real telecom protocols
- Could not be blocked without harming real customers

4.4 Stage 4: Impact Attempt

If the attack had succeeded, it could have caused:

- Loss of mobile service in parts of New York City
- Failure of 911 emergency calls
- Problems with financial authentication systems
- Disruption of IoT devices such as traffic systems, alarms, and logistics units
- Service degradation for millions of users

Telecom infrastructure is interconnected. A regional failure could spread across states.

5. Impact Analysis

5.1 Direct Impact (Potential)

- Telecommunications outage
- Large scale service disruption
- Overload of emergency call routing
- Delays in police and fire communications
- Public safety risks

5.2 Critical Infrastructure Impact

New York City depends heavily on mobile networks for:

- Hospitals
- Financial services and Wall Street
- Transportation
- Fire, police, and emergency units
- Government communication

Any disruption could cause cascading failures.

5.3 National Security Impact

A coordinated attack on U.S. communications is considered:

- A national security threat
- A possible early stage of a larger cyber operation

- A form of asymmetric warfare

6. Why the Attack Was So Dangerous

- It bypassed traditional DDoS defences because the traffic came from mobile networks.
- It targeted the control plane, which has limited capacity.
- It used massive scale through SIM farms.
- It exploited global telecom trust relationships.
- It could cause serious damage during a crisis, even if it lasted only minutes.

7. Failure Points in Defence

- Weak rate limits for signaling traffic
- Outdated SS7 and Diameter protections
- Lack of anomaly detection for authentication storms
- Poor separation between control plane and data plane resources
- Excessive trust in foreign carriers
- Inadequate monitoring of SIM identity anomalies

8. Countermeasures and Recommendations

8.1 Strengthen Signaling Security

- Improve SS7 and Diameter firewalls
- Add rate limiting based on subscriber groups
- Use AI based anomaly detection for signaling floods

8.2 SIM Identity Protections

- Real time IMSI and IMEI mismatch detection
- SIM cloning detection
- Device fingerprinting for SIM cards

8.3 Network Hardening

- Segmentation of authentication servers
- Redundant signaling paths
- Load balancing for attach requests

8.4 Threat Intelligence Sharing

Telecom providers should share:

- Attack patterns
- SIM device signatures
- Traffic anomalies
- Information on foreign origin signaling threats

8.5 Government and Industry Collaboration

A national level response plan should involve:

- DHS
- FCC
- NSA
- CISA
- Telecom providers

9. Lessons Learned

- Telecom networks have hidden vulnerabilities.
- DDoS attacks are evolving to target signaling systems instead of IP networks.
- Nation state actors examine U.S. infrastructure for weak points.
- Mobile networks generate traffic that looks legitimate, making attacks hard to block.
- Large cities like New York need advanced telecom security controls.

10. Conclusion

The SIM based DDoS attack described by researchers represents a new style of cyber threat targeting the core of U.S. communication systems. Instead of malware or brute force attacks, this method uses signaling floods at the telecom level. It is highly sophisticated, extremely difficult to detect, and potentially capable of disabling critical areas of a major city.

The incident shows that the United States must rapidly improve telecom security to protect vital infrastructure. Without stronger defences, adversaries can launch cyber operations that cause real world consequences affecting millions of people.